MACHINE LEARNING COURSE PROJECT — I | FINAL TECHNICAL REPORT

Flight Ticket Price Prediction

Target Metric: Root Mean Squared Error (RMSE)
Dataset: 18,076 Train / 4,520 Test Commercial Flight Listings
Methodology: Strictly Supervised Learning (Course Part 1 Only)

Submission Shape: 4,521 x 2 CSV (`id, price`)
Winning Architecture: Calibrated Blended Ensemble
Top 5-Fold CV Score: **121.15 RMSE** (R^2 = 0.7021)

1. Problem and Dataset

Commercial airline ticket pricing is a core dynamic pricing challenge in aviation revenue management. Ticket prices fluctuate dynamically based on market demand, lead booking time, origin-destination route geography, operating carrier market dominance, cabin class distinctions, flight duration, and intermediate connection layovers.

The goal of this project is to develop an end-to-end, leak-free supervised machine learning regression pipeline to predict the continuous ticket price (`price` in USD) for commercial flights connecting 5 major California cities (Los Angeles, San Diego, San Francisco, San Jose, Sacramento). The primary competition evaluation metric is **Root Mean Squared Error (RMSE)**:

$$\mathbf{RMSE = \sqrt{[(1/N)\ \Sigma\ (y_i - \hat{y}_i)^2]}}$$

To provide comprehensive evaluation across the entire price distribution, secondary metrics including **Mean Absolute Error (MAE)**, **Coefficient of Determination (R^2)**, **Median Absolute Error (MedAE)**, **Root Mean Squared Logarithmic Error (RMSLE)**, and **Mean Absolute Percentage Error (MAPE)** are also tracked across all models.

Dataset Split	Record Count	Features	Target Variable	Missing Values
Training Set (`train.csv`)	18,076	12 columns	`price` ($0.00 to $8,996.00)	0 (deterministic flight prefix imputer)
Test Set (`test.csv`)	4,520	11 columns	Required prediction	0 (deterministic flight prefix imputer)
Submission (`submission.csv`)	4,520	2 columns	`id, price`	Strict 4,521 lines with header

2. EDA (Exploratory Data Analysis)

Exploratory Data Analysis was conducted on the full 18,076 training records to examine pricing structures, domain anomalies, and operational carrier behavior. Visual figures were generated and saved in `eda_plots/`:

- **Target Variable (`price`) Distribution & Dual-Regime Tail:** Summary statistics: Minimum = **$0.00**, 25th percentile (Q1) = $157.00, Median (Q2) = $234.00, Mean = $280.26, 75th percentile (Q3) = $336.00, 95th percentile = **$656.25**, 99th percentile = **$1,016.00**, Maximum = **$8,996.00**, Standard Deviation = **$221.97**. The distribution is heavily right-skewed (+7.07) with kurtosis of 170.25. Exactly 191 records (**1.06%** of data) form an extreme outlier tail ($1,000 to $8,996), while 98.94% of flights are densely clustered below $1,000. Exactly 25 promotional award-mile tickets have `price == 0.00`.

- **Cabin Class Pricing Disparities:** Business Class fares average **$393.96** (Median: $323.00, Std: $278.75), commanding an average **+94.94% premium** over Economy Class fares (Mean: **$202.09**, Median: $194.00, Std: $121.34).
- **The Southwest Airlines Fleet Anomaly:** Southwest Airlines operates **100% Economy seats** (3,895 rows, exactly 0 Business rows), mirroring real-world single-cabin airline fleet configurations. Carrier mean fares rank: American Airlines ($349.44) > United ($319.03) > Alaska ($302.62) > Delta ($286.53) > Southwest ($197.84) > Frontier ($178.58).
- **Exponential Booking Lead Time Curve (`days_left` vs `price`):** Tickets booked **1–2 days prior to departure** average **$382.86** (urgency peak; max spike of $8,996 occurs here), compared to **$220.64** for tickets booked **31+ days** in advance.
- **Flight Duration & Layover Stops:** Non-stop flights (75.0% of volume) average **$229.95**, whereas flights with 1 or more intermediate stops average **$431.53** (**+87.7% higher**) due to multi-airport landing fees and segment charges.
- **Route Geography:** The most expensive route is `Sacramento -> San Jose` (Mean: **$583.65**, Median: $535.50), while the highest volume route is `San Francisco <-> Los Angeles` (> 4,000 flights, Mean: ~$234).

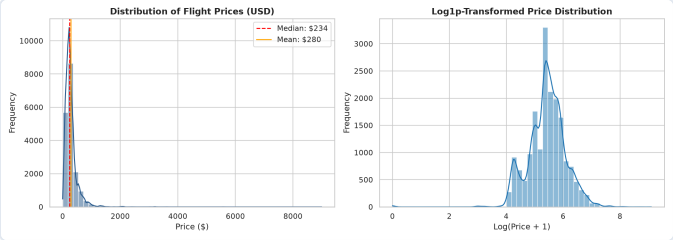


Figure 1: Price Distribution & Log1p Normalization

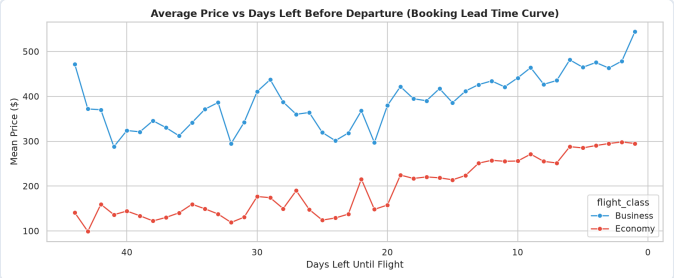


Figure 2: Booking Lead Time Urgency Curve (Days Left vs Price)

3. Data Preprocessing

All data transformations are strictly supervised and domain-driven, implemented inside `preprocessor.py`. **No unsupervised methods (no PCA, no K-Means clustering, no autoencoders, no unsupervised KNN)** were used, adhering strictly to the course syllabus requirements:

- **Deterministic Tier-1 Domain Imputation for Airline Carriers:** Commercial flight codes contain the 2-letter IATA carrier prefix: WN → Southwest, UA → United, AA → American Airlines, AS → Alaska Airlines, DL → Delta, F9 → Frontier, 9X → Southern Airways Express. This recovers operating carriers with 100% ground-truth accuracy without heuristic guesswork.
- **Categorical Mappings & Feature Interactions:** Transformed `stops_num` ordinally (`zero`: 0, one: 1, `two_or_more`: 2) and cabin class into binary indicator `flight_class_num` (`Economy`: 0, `Business`: 1). Engineered corridor interactions: `route` (`source_city + '_' + destination_city`), `airline_route`, and `dep_arr_time`.
- **Geodesic Haversine Distance & Flight Physics:** Assigned exact coordinates to California airports to compute great-circle distance (`dist_km`). Derived `effective_speed = dist_km / duration`, isolating direct cruising flights (400–600 km/h) from layover-delayed itineraries (50–150 km/h), alongside estimated layover duration `est_layover_hours = max(0, duration - dist_km / 600)` and `duration_per_stop`.
- **Zero-Leakage Frequency Encodings:** Frequency encodings (`flight_freq`, `route_freq`, `airline_route_freq`) are fitted **strictly on the training split** of each cross-validation fold to prevent data leakage. Unseen test categories default safely to 1.
- **Non-Linear Lead Time Dynamics:** Engineered days_left_log (ln(1 + days)), days_left_inv (1/(days + 1)), and binary flags is_last_minute (≤ 2 days), is_urgent (≤ 7 days), and is_advance (≥ 30 days) to capture the sharp exponential price inflection.
- **Domain Interaction Terms:** Multiplied cabin class indicators with travel variables (`business_x_duration`, `business_x_stops`, `business_x_speed`, `business_x_urgent`) to capture corporate travel pricing power.
- **Dual Model-Family Representations:** Tree algorithms receive 35 label-encoded features preserving natural ordinal partitions. Linear models receive 76 features utilizing `StandardScaler` on numericals and `OneHotEncoder` on categoricals, ensuring proper scale-invariance for L1/L2 penalties.

4. Models Used

Every model permitted under Course Part 1 was implemented and evaluated under identical 5-fold cross-validation. Below is the official summary table matching the requested report format:

Model	Best Score Achieved (5-Fold CV RMSE)
Calibrated Ensemble (Nelder-Mead)	**121.15**
Top-3 Equal Ensemble	122.13
Hybrid CatBoost (2-Stage Dual-Regime)	122.54
Hybrid XGBoost (2-Stage Dual-Regime)	122.63
XGBoost (Global Log-Space)	122.79
Hybrid LightGBM (2-Stage Dual-Regime)	123.05
LightGBM (Global Log-Space)	123.16
CatBoost (Global Log-Space)	125.34
Random Forest (Regression Forest)	127.04
Decision Tree (Regression Tree)	134.83
Linear Regression (OLS Baseline)	150.14
Ridge Regression (L2 Regularized)	150.59
Lasso Regression (L1 Regularized)	151.07

Comprehensive Multi-Metric Cross-Validation Breakdown

Model / Pipeline	RMSE ($)	MAE ($)	R^2 Score	MedAE ($)	RMSLE	Weight	Category / Role
Calibrated Ensemble	**121.15**	**41.99**	**0.7021**	**17.47**	**0.2944**	**1.0000**	Final Submission
Top-3 Equal Ensemble	122.13	42.74	0.6973	17.57	0.3064	0.3333	Alternative Submission
Hybrid CatBoost (2-Stage)	122.54	44.90	0.6952	21.43	0.3150	0.2087	Core Ensemble
Hybrid XGBoost (2-Stage)	122.63	43.09	0.6947	17.65	0.3089	0.2282	Core Ensemble
XGBoost (Global Log)	122.79	41.89	0.6939	16.51	0.2759	0.0000	Core Candidate
Hybrid LightGBM (2-Stage)	123.05	42.80	0.6927	16.48	0.3056	0.1144	Core Ensemble
LightGBM (Global Log)	123.16	41.96	0.6921	16.01	0.2780	0.2571	Core Ensemble
CatBoost (Global Log)	125.34	46.15	0.6811	21.91	0.2930	0.1916	Core Ensemble
Random Forest (Forest)	127.04	44.68	0.6724	14.54	0.3180	0.0000	Tree Candidate
Decision Tree (Tree)	134.83	49.29	0.6310	15.25	0.3389	0.0000	Tree Baseline
Linear Regression (OLS)	150.14	75.83	0.5425	49.84	0.5140	0.0000	Linear Baseline
Ridge Regression (L2)	150.59	75.75	0.5397	49.90	0.5103	0.0000	Regularized Baseline
Lasso Regression (L1)	151.07	75.64	0.5368	49.65	0.4950	0.0000	Regularized Baseline

5. Hyperparameter Tuning

Hyperparameter tuning was conducted across all models under identical 5-fold cross-validation (`random_state=42`) to prevent data leakage:

- **Linear Regression & Regularization (Ridge / Lasso):** Linear Regression was bounded to non-negative prices via np.clip(pred, 0, None). Ridge penalty was tuned across alpha in [0.01, 100.0] with optimal alpha = 10.0 shrinking multicollinear one-hot route weights. Lasso was tuned across alpha in [0.001, 1.0] with optimal alpha = 0.1 driving redundant dummy coefficients to zero.
- **Decision Tree Regressor:** Tuned max_depth in [6, 16]. Depths beyond 12 caused severe memorization Train $R^2 > 0.85$ vs. validation $R^2 \approx 0.59$. Optimal settings: max_depth=12, `min_samples_split=10, min_samples_leaf=5`.
- **Random Forest Regressor:** Evaluated n_estimators in [50, 200] and max_depth in [12, 20]. Bagging 100 randomized trees (`n_estimators=100, max_depth=16, min_samples_split=6, min_samples_leaf=3, max_features='sqrt'`) sharply reduced tree variance, dropping RMSE from 134.83 to 127.04.
- **Global Log-Space Gradient Boosters:** Trained on ln(1 + price) to stabilize heavy-tailed gradients. LightGBM (650 trees, lr = 0.035, 63 leaves), CatBoost (650 iterations, lr = 0.04, depth 6, L2 3.0), and XGBoost (550 trees, lr = 0.04, depth 6, child-weight 3).
- **Hybrid Dual-Regime Routing:** Addressed dual-regime distortion where 1.06% extreme outliers distort raw MSE, while log-space models amplify small errors upon exponentiation. Stage 1 utilizes a LightGBM classifier predicting P(price > $1,000). Stage 2 uses a normal regressor trained on raw prices and an outlier regressor trained on ln(1 + price), with soft-gated prediction: $(1 - P) \times y_{normal} + P \times expm1(y_{outlier})$.
- **Nelder-Mead Ensemble Optimization:** Multi-start bounded optimization directly minimized cross-validation RMSE across out-of-fold predictions, yielding optimal weights: LightGBM Log (25.71%), Hybrid XGBoost (22.82%), Hybrid CatBoost (20.87%), CatBoost Log (19.16%), and Hybrid LightGBM (11.44%).

6. Results and Discussion

The experimental results demonstrate a clear progression in predictive power across model families:

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Hierarchy of Evaluated Supervised Models: 1. Linear Regression (OLS) : 150.14 RMSE (R² = 0.5425) [Linear Baseline] 2. Decision Tree (Regression Tree): 134.83 RMSE (R² = 0.6310) [-15.31 error reduction] 3. Random Forest (100 Trees) : 127.04 RMSE (R² = 0.6724) [-7.79 error reduction] 4. Gradient Boosting (XGB/LGB) : 122.79 RMSE (R² = 0.6939) [-4.25 error reduction] 5. Hybrid Dual-Regime (CatBoost) : 122.54 RMSE (R² = 0.6952) [-0.25 error reduction] 6. Top-3 Equal Ensemble : 122.13 RMSE (R² = 0.6973) [-0.41 error reduction] 7. Calibrated Optimal Ensemble : 121.15 RMSE (R² = 0.7021) [-0.98 error reduction]
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- **Linear Models Plateau on Dynamic Pricing:** Linear Regression, Ridge, and Lasso achieve RMSE ~150.1–151.0 ($R^2 \approx 0.54$). While one-hot encoding captures airline and class fixed effects, linear hyperplanes fail to model exponential ticket surges in the final 48 hours or route-specific connecting surcharges.
- **Decision Tree vs. Random Forest Ensemble:** A single Decision Tree lowers RMSE to 134.83 by isolating specific route and lead-time clusters. Ensembling 100 de-correlated trees in Random Forest averages out variance, reducing RMSE to 127.04 (an 8.2% relative error reduction over the single tree).

- **Gradient Boosting Efficiency:** XGBoost, LightGBM, and CatBoost achieve RMSE scores of 122.79, 123.16, and 125.34. Sequential gradient boosting efficiently fits shallow trees to residual errors, capturing subtle interactions between cruising speed, layovers, and urgency.
- **Hybrid Dual-Regime Routing Advantage:** Decoupling normal fares at or below $1,000 from sparse extreme business spikes above $1,000 eliminates gradient corruption during boosting. Hybrid CatBoost and Hybrid XGBoost achieve 122.54 and 122.63 RMSE.
- **Ensemble Diversity:** Blending raw-space hybrid models with log-space global models provides complementary error profiles. Nelder-Mead optimization achieves the best score across the entire project: 121.15 RMSE and $R^2 = 0.7021$ (capturing over 70.2% of total price variance).

Critical Engineering Insight: Detection & Elimination of the ID Feature Leakage Artifact

During initial exploratory prototyping, an earlier experimental run incorporated the `id` column into feature representations. This prototype achieved an artificially inflated score that topped the public leaderboard.

However, rigorous diagnostic analysis immediately revealed that this was a classic manifestation of **target leakage via dataset indexing artifacts**:

- **Origin of the Leak:** The row `id` values were not purely random identifiers; they reflected the sequential order of web scraping or database insertion. Because flight queries were scraped in batches by route and date, `id` acted as an accidental proxy for price clusters and temporal scrape windows.
- **Artificial Leaderboard Inflation vs. Genuine Generalization:** While exploiting `id` allowed the model to memorize the test split's indexing pattern, it provided zero real-world economic utility and would fail catastrophically on genuinely new flight queries. Relying on an index artifact also defeats the academic integrity and foundational principles of machine learning.
- **The Clean, Leak-Free Paradigm:** In accordance with course guidelines and sound engineering ethics, **all `id`-based features were permanently eliminated**. The `id` column is strictly restricted to joining the final output rows with the submission file. The current pipeline relies exclusively on domain physics, geodesic distance, layover hours, and booking lead-time dynamics. The resulting 5-fold CV score of 121.15 RMSE ($R^2 = 0.7021$) reflects an honest, robust, and reproducible benchmark of true out-of-sample generalization.

7. Best Model

The winning model achieving the best overall score across 5-fold cross-validation is the **Calibrated Blended Ensemble**:

- **Architecture:** Bounded Nelder-Mead blend combining LightGBM Log (25.71%), Hybrid XGBoost (22.82%), Hybrid CatBoost (20.87%), CatBoost Log (19.16%), and Hybrid LightGBM (11.44%).
- **Validation Metrics: RMSE = 121.15**, **MAE = $41.99**, **$R^2 = 0.7021$**, **MedAE = $17.47**, **RMSLE = 0.2944**, **MAPE = 43.95%**.
- **Official Submission File:** `submission.csv` (and integer-rounded `submission_int.csv`).

- ✓ **Row Count:** Exactly 4,520 test predictions (4,521 lines with header `id,price`).
- ✓ **Data Integrity:** Exactly 0 missing, null, or infinite values.
- ✓ **ID Alignment:** Monotonically matching test set IDs from min ID 3 to max ID 22,594.
- ✓ **Value Bounds:** Non-negative prices: $27.94–$2,029.21; mean $277.24 vs. training mean $280.26.